

GLAiCemic: An AI-Based Food Image Recognition, Nutritional Estimation, and Predictive Analysis System for Diabetic Patients

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I. PROPOSED TITLE

GLAiCemic: An AI-Based Food Image Recognition, Nutritional Estimation, and Predictive Analysis System for Diabetic Patients

II. BACKGROUND OF THE STUDY

Diabetes mellitus and other non-communicable diseases (NCDs) continue to pose significant public health challenges in the Philippines. According to the Philippine Statistics Authority (PSA), diabetes mellitus was the fifth leading cause of death in the country in 2024, accounting for 43,989 deaths. During the same year, ischemic heart disease remained the leading cause of death with 134,074 recorded deaths, while hypertensive diseases accounted for 38,088 deaths. These conditions are associated with several modifiable risk factors, including unhealthy dietary practices, physical inactivity, overweight, and obesity. The continuing burden of these diseases emphasizes the importance of preventive healthcare strategies that support individuals in managing their lifestyle, nutrition, and other health-related behaviors.

Proper nutrition plays an important role in the prevention and management of diabetes and other diet-related non-communicable diseases. Findings from the 2023 National Nutrition Survey (NNS) conducted by the Department of Science and Technology Food and Nutrition Research Institute (DOST-FNRI) further demonstrate the commonness of nutrition-related health concerns among Filipino adults. The survey reported substantial levels of overweight and obesity among adults and identified elevated blood pressure as another continuing health concern, particularly among older age groups. Furthermore, the World Health Organization (WHO) reported that non-communicable diseases accounted for approximately 68% of deaths in the Philippines in 2021. These findings highlight the importance of promoting healthier food choices, improving nutritional awareness, and providing accessible tools that can assist individuals, particularly those living with chronic conditions, in monitoring and managing their daily dietary intake.

At the local level, diabetes mellitus is also a significant health concern in the Municipality of San Rafael, Bulacan. Based on the 2024 Annual Field Health Services Information System (FHSIS) Report of the Department of Health Central Luzon Center for Health Development, a total of 2,017 individuals with diabetes mellitus were identified in San Rafael in 2024. Among these individuals, 1,816, or approximately 90%, were provided with medicines for the management of their condition. The presence of more than two thousand identified individuals with diabetes within the municipality demonstrates the need for continuous disease management

and accessible interventions that can complement existing healthcare services, particularly those that can assist patients in monitoring factors associated with diabetes management, including their daily dietary intake.

The Rural Health Unit II (RHU II) of San Rafael, Bulacan is a government primary healthcare facility under the Municipal Health Office of San Rafael. Located at Maguinao DRT Highway San Rafael, Bulacan, it is one of the Rural Health Units in San Rafael that is responsible for delivering accessible primary healthcare services to residents of the San Rafael. Operating under the supervision of Department of Health (DOH) and the Local Government Unit (LGU) of San Rafael, RHU II provides essential healthcare services, including medical consultations, maternal and childcare, immunization, nutrition programs, health education, and the management of communicable and non-communicable diseases. Headed by Dr. Ma. Josefina Editha Sadam, the Municipal Health Office oversees the implementation of healthcare programs and ensures delivery of quality primary healthcare services. RHU II also utilizes the Department of Health (DOH) health information system to manage patient records, monitor health programs, and improve the efficiency of healthcare service delivery, supporting its mission to promote preventive healthcare and improve the well-being of the residents of San Rafael, Bulacan.

During the data gathering, the researchers did an interview on one of the head personnel of the Rural Health Unit II (RHU II) of San Rafael, Bulacan, based on the gathered data that many patients are suffers from chronic illnesses such as diabetes, heart disease, high blood pressure and other diseases which requiring them to carefully monitor their food intake where nutritional components can be presents such as sugar, carbohydrates, calories, and sodium. Although the RHU personnels already provides nutritional counseling during consultations, many patients still experiencing some problem in identifying the right food to take specially when taking daily meals making it challenging to consistently follow recommended dietary guidelines and effectively manage their health conditions.

The Rural Health Unit II currently utilizes the YAKAP Application, which a Department of Health (DOH) supported digital health application which features patient registrations, health record management, and monitoring of health care services. However, this is only primarily designed for patient information management and other healthcare services and does not provide patient-centered features such as AI-powered food recognition, nutritional estimation, or daily meal monitoring for patients. This creates a gap between the healthcare services provided during consultations and the continuous dietary monitoring needed by patients in their daily lives. To address this gap, the researchers proposed GLAiCemic: An AI-Based Food Image Recognition, Nutritional Estimation, and Predictive Analysis System for Diabetic Patients.

The proposed system will integrate Artificial Intelligence (AI) to determine or recognize food from the uploaded or captured images, and it will automatically calculate and show the nutritional values, including sugar, calories, carbohydrates, protein, fat, fiber, and sodium. This will also serve as predicative analysis of diabetes and other possible health risks based on the

gather nutritional values. Rather than replacing the YAKAP App provided by the Department of Health (DOH), GLAiCemic is designed to with additional features that can provide diabetic patients with accessible tools for monitoring their daily food intake, dietary planning, monitoring of their diet, and improving self-management for their health.

III. PROBLEM STATEMENT

General Problem

This study aims to determine how an AI-based food image recognition and nutritional estimation system can enhance dietary monitoring, nutritional assessment, and diabetes self-management among diabetic patients through Artificial Intelligence, personalized nutritional recommendations, and predictive analysis.

Specific Problems

The proposed study seeks to answer the following questions:

Specific Problems

The proposed study seeks to answer the following questions:

1. How can Artificial Intelligence (AI) be utilized to accurately recognized food items from captured and uploaded images.
2. How can the system enable diabetic patients to record, monitor, and review, their daily meals and blood glucose levels to support effective diabetes self-management?
3. How can the system estimate the nutritional content of recognized food items, including calories, carbohydrates, sugar, protein, fat, fiber, and sodium, to assist diabetic patients in making informed dietary decisions?
4. How can the system provide personalized nutritional recommendations to promote healthier eating habits and improve diabetes self-management?
5. How can predictive analysis be utilized to analyze diabetic patients meal history, nutritional intake, and blood glucose records in predicting potential blood glucose trends and dietary risks?

IV. OBJECTIVES

General Objectives

The researchers aims to develop a mobile and web-based application that utilizes Artificial Intelligence to recognize food items, estimate nutritional content, monitor dietary intake and blood glucose levels, and generate predictive insights to support effective diabetes self-management.

Specific Objectives

- To develop an mobile based AI food image recognition module that accurately identifies food items from captured or uploaded images.
- To implement a nutritional estimation feature that calculates the nutritional content of recognized food items, including calories, carbohydrates, sugar, protein, fat, fiber, and sodium.
- To provide personalized nutritional insights and dietary recommendations based on the recognized food and estimated nutritional values to promote healthier food choices and better diabetes self-management.
- To design and develop a meal logging, nutritional monitoring, and blood glucose monitoring module that enables diabetic patients to record, track, and review their daily dietary intake and blood glucose levels.
- To develop a predictive analysis module that analyzes users' meal history, nutritional intake, and blood glucose records to forecast blood glucose trends and identify potential dietary risks, enabling proactive diabetes self-management.

V. TARGET USERS

Primary Users

- **Diabetic Patients.** The primary users who will utilize the system to recognize food through AI, estimate nutritional values, log meals and blood glucose levels, receive predictive insights, and monitor their overall dietary habits for better diabetes self-management.
- **Rural Health Unit Administrators** (Doctors, Nurses, and Nutritionists/Dietitians). They can review patients' nutritional records, blood glucose history, and predictive reports to provide personalized dietary counseling and monitor patient progress.

Secondary Users

- **Municipal Health Office (MHO).** Authorized personnel may access aggregated and anonymized reports to monitor diabetes-related nutrition trends, evaluate community health programs, and support data-driven planning for non-communicable disease (NCD) initiatives.
- **Non-Diabetic Individuals.** Individuals without diabetes may use GLAiCemic to analyze the nutritional content of their meals, monitor their daily dietary intake, and develop healthier eating habits. The system's AI-powered food recognition and nutritional estimation features can also help users become more aware of their nutritional consumption and encourage preventive health practices to reduce the risk of developing lifestyle-related diseases such as diabetes and hypertension.

VI. PROPOSED TECHNOLOGIES

During the development of the proposed system, the researchers will utilize the following tools that are widely available:

Software Development Tools

- Visual Studio Code
- Git & GitHub
- PostMan
- XAMPP
- Stitch and Figma

Programming Languages

- Python
- PHP
- JavaScript
- SQL

Frameworks

Web-based System

- Django
- Bulma
- Chart.js
- HTML5

Mobile Application

- React Native (Expo)

AI Backend

- FastAPI

Database

- MySQL

Artificial Intelligence Technologies

- YOLOv8
- Ultralytics
- OpenCV

- XGBoost
- PyTorch

Hardware

1. Development

- Laptop/Desktop (Intel Core i5/Ryzen 5 or higher)
 - Windows 10 (64-bit) or Windows 11
 - Storage: Minimum **256 GB SSD** (512 GB recommended)
 - Minimum **8 GB RAM** (16 GB recommended)
- Webcam
- Android smartphone
 - Camera for food image scanning
 - **Operating System:** Android 10 or later
 - Minimum 4 GB RAM
 - Stable internet connection

2. Deployment

- Android Smartphone
 - Operating System: Android 10 or later
 - Camera
 - Stable internet connection
- Desktop/Laptop
 - Operating System: Windows 10 (64-bit) or Windows 11
 - Modern web browser:
 - Google Chrome (latest version)
 - Microsoft Edge (latest version)
 - Stable internet connection

VII. INNOVATION

GLAiCemic was developed to provide its own uniqueness by integrating multiple core features such as AI food image recognition, automatic nutritional estimation, and predictive analysis into a single platform designed for diabetic patients and even for the community. Compared to other existing nutrition applications such as MyFitnessPal, Foodvisor, CalorieMama, YAZIO, and Lose it! provides calorie counting, nutritional tracking, and some cases already AI-assisted. However, these applications are only designed for general fitness, weight management, or healthy lifestyle monitoring and not specifically focused on dietary management needs for diabetic patients within a community healthcare setting. It is also generally operating as standalone consumer applications and are not integrated with local healthcare providers for patient monitoring.

GLAiCemic has the capability to analyze the user's meal history, nutritional intake, and blood glucose records to identify the dietary patterns and generate the predictive insights that can support proactive diabetes self-management. Additionally, the system also provides dual-platform architecture, which consists of mobile application that can be used for diabetic and non-diabetic users to monitor their health status and web application for the Rural Health Unit administrators and healthcare personnels for monitoring, reviewing, and analyzing for generating reports. By complementing the existing **YAKAP App** used by RHU II, GLAiCemic extends healthcare support beyond clinical consultations and promotes continuous patient monitoring, making it a more comprehensive and community-oriented solution for diabetes management.

VIII. FEASIBILITY

Technical Feasibility

The proposed system is technically feasible because it will utilize free and widely available technologies such as PHP/Django, React Native (Expo), Python, YOLOv8, PyTorch, MySQL, and Bulma. These technologies support AI integration, cross-platform development, and database management while minimizing development costs. The researchers possess the necessary knowledge and skills in software development and AI implementation to develop the proposed system.

Operational Feasibility

The system is operationally feasible as it is designed to support daily monitoring and diabetes self-management needs for a diabetic patients by accessing the mobile application through their mobile phone, while providing additional tools for healthcare professionals and Rural Health Units with a web-based platform for monitoring patient nutritional records, patients medical records, and predictive analysis reports. The system is user-friendly interface and automated AI-powered features reduce the manual logging and nutritional computation, making it easier for patients to monitor their health and for healthcare providers to track patient progress, suggesting nutritional guidance, and facilitating informed decision-making during consultations.

Economic Feasibility

The proposed system is also economically feasible since it utilizes free and open-source software, frameworks, libraries, and development tools, which can reduce the development and implementation cost. The only significant expenses involved is internet connectivity, domain name, cloud hosting services and other tools used for development and testing.

Schedule Feasibility

The development of the system can be completed within the allotted capstone development period. The proposed schedule includes planning, requirements gathering, system designing, development, testing, and deployment. The researchers selected Rural Health Unit II (RHU II) of

San Rafael, Bulacan as the partner institution because it is readily accessible, allowing for efficient coordination, regular consultations, and timely collection of requirements and feedback throughout the development process. The accessibility and cooperation of the target institution facilitate system validation, user testing, and evaluation with the intended users. Furthermore, the availability of existing AI models and open-source technologies helps reduce development time, contributing to the successful completion of the project within the planned schedule.

IX. SCOPE AND LIMITATIONS

The proposed system, GLAiCemic: An AI-Based Food Image Recognition, Nutritional Estimation, and Predictive Analysis System for Diabetic Patients, is a web- and mobile-based healthcare system designed to assist individuals with Type 1 diabetes, Type 2 diabetes, gestational diabetes, and prediabetes in monitoring their nutritional intake and supporting diabetes self-management. The mobile application allows users to capture or upload food images for AI-based food recognition, automatically estimate nutritional values such as calories, carbohydrates, sugar, protein, fat, fiber, and sodium, record daily meals and blood glucose levels, and receive predictive insights based on their meal history, nutritional intake, and blood glucose records. The web-based application is intended for Rural Health Unit administrators and authorized healthcare professionals for managing patient accounts and medical records, monitoring patients' nutritional intake, reviewing predictive analysis reports, and generating patient monitoring reports to support healthcare consultations and decision-making. The system focuses on nutritional monitoring, glucose tracking, and predictive insights for the identified target groups and is intended to serve as a supportive healthcare tool rather than a replacement for professional medical diagnosis or treatment.

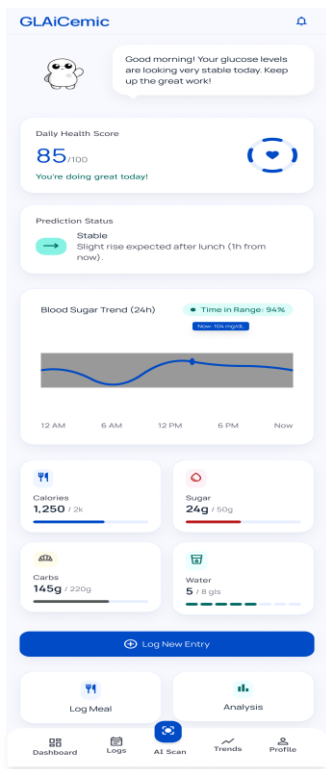
The study acknowledges several limitations, particularly in recognizing food items that are included in its trained AI dataset, and the accuracy of nutritional estimations depends on the quality of the captured images and available nutritional database. The predictive analysis provides forecasts and insights based on users' historical meals and blood glucose records and is intended solely as a decision-support tool; it does not provide medical diagnoses or replace professional medical advice. Additionally, the system requires an internet connection for AI processing and data synchronization, and it is designed to complement existing healthcare practices rather than replace healthcare professionals or official health information systems used by the Rural Health Unit.

X. EXPECTED OUTPUT

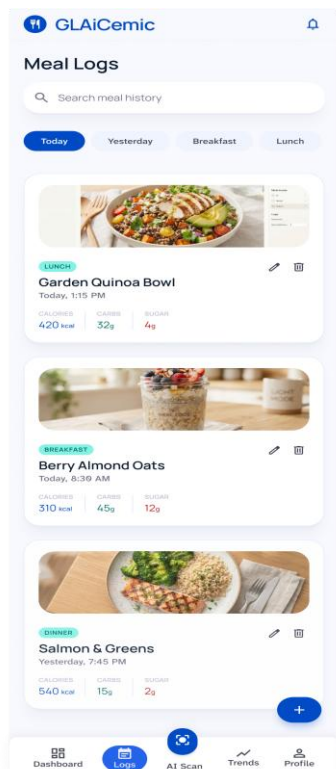
The expected output of this study is **GLAiCemic: An AI-Based Food Image Recognition and Nutritional Estimation System for Diabetic Patients**, consisting of a **mobile application** for food scanning, nutritional estimation, meal and blood glucose monitoring, and predictive analysis, and a **web-based dashboard** for RHU administrators and healthcare professionals to monitor patient records and generate reports. The project will also deliver a trained AI model, a

centralized MySQL database, and complete system documentation to support diabetes self-management and patient monitoring.

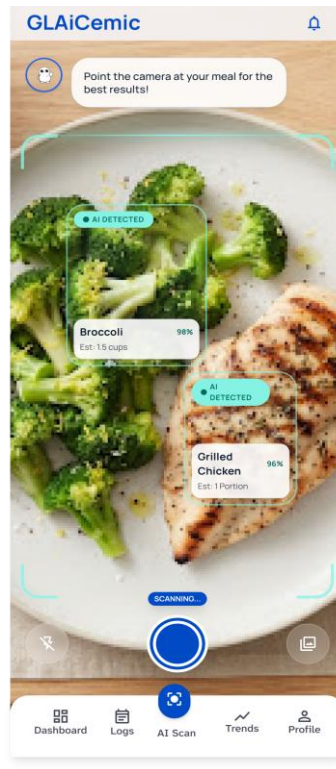
Mobile Application Prototype



Health Dashboard



Meal Logging Module

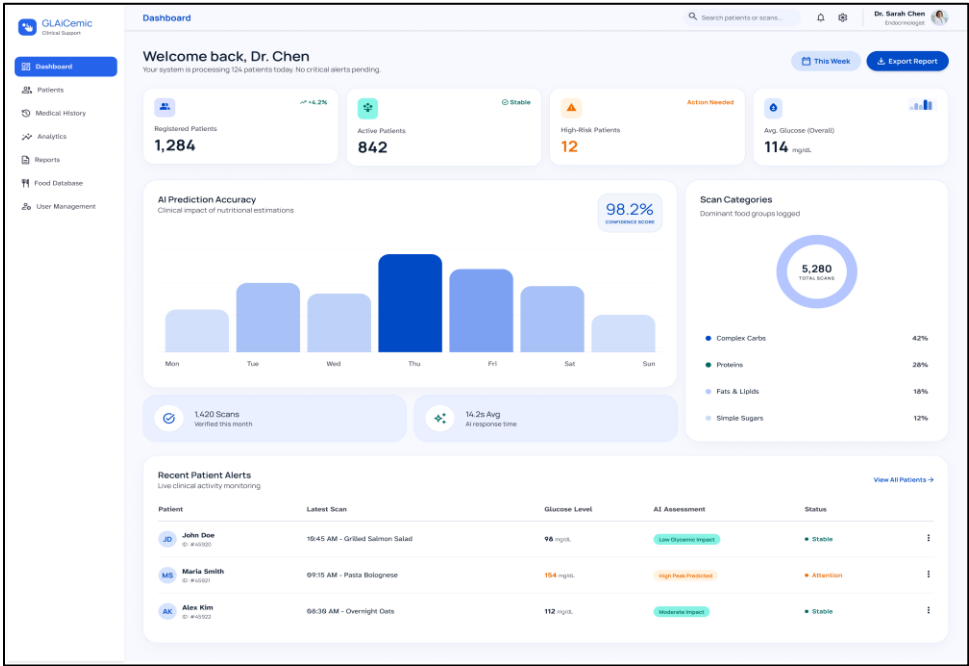


AI Food Recognition

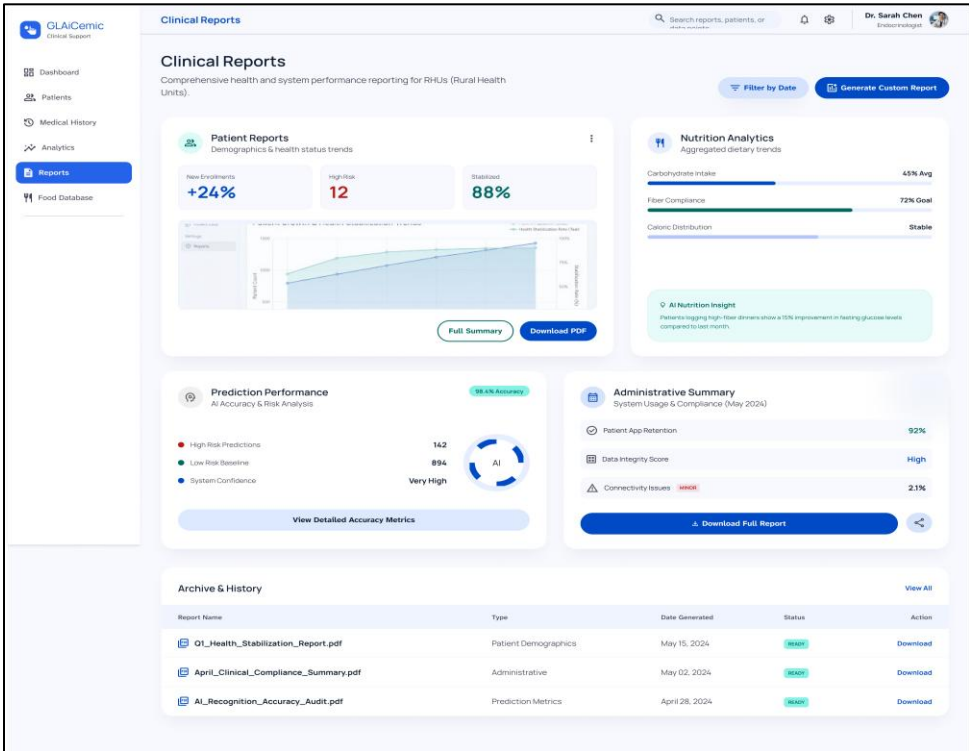


Nutritional Estimation

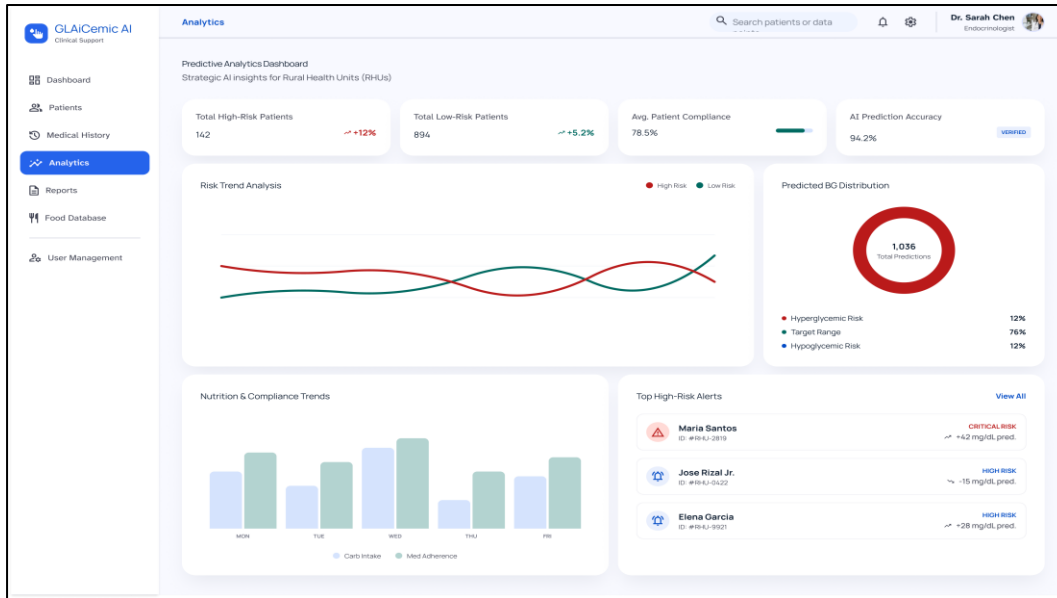
Web Application Prototype



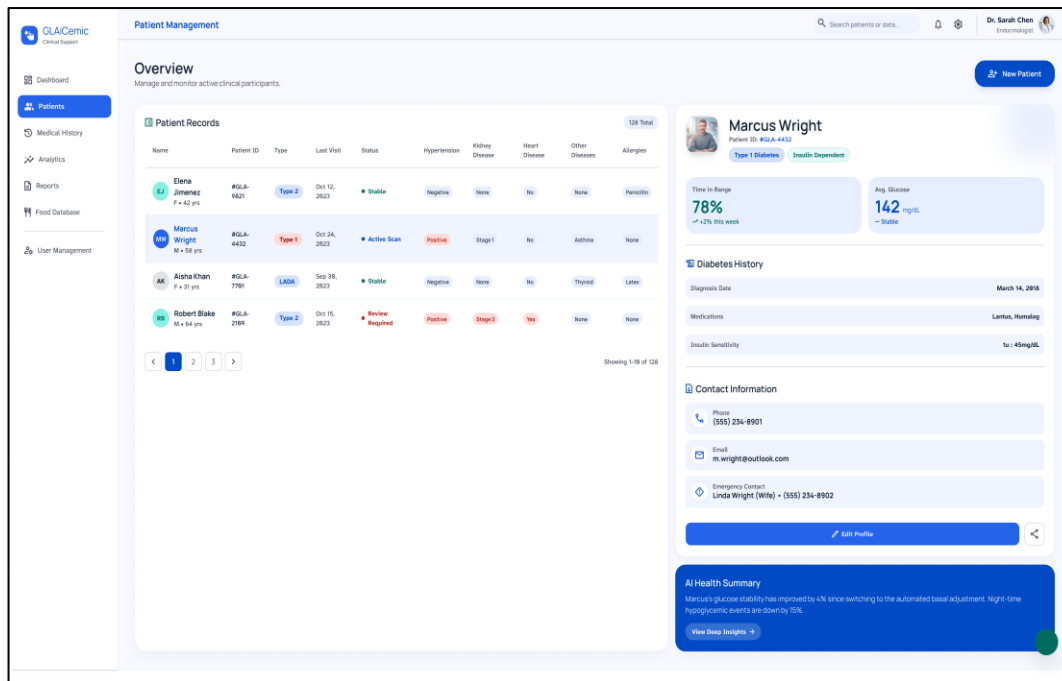
Web Dashboard



Reports Module



Predictive Analytics Module



Patient Management Module

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